

LAURINBURG-MAXTON, NC, USA

WMO: 723109

Lat: 34.792N

Lon: 79.366W

Elev: 67

StdP: 100.52

Time Zone: -5.00 (NAE)

Period: 99-19

WBAN: 93782

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -6.0	(c) -3.9	(d) -15.5	(e) 1.0	(f) -1.6	(g) -12.8	(h) 1.3	(i) 0.7	(j) 10.4	(k) 11.6	(l) 9.1	(m) 10.1	(n) 1.7	(o) 50	(p) 0.421

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.7	35.5	24.3	34.0	24.1	32.9	23.8	26.3	32.0	25.7	31.2	25.2	30.4	3.7	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.8	20.1	28.4	24.1	19.2	27.7	23.7	18.7	27.3	81.8	32.1	79.0	31.3	76.8	30.5	30.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8.7	7.6	6.9	-9.6	37.3	2.3	1.5	-11.3	38.4	-12.6	39.3	-13.9	40.1	-15.6	41.2
			-10.6	27.4	2.1	1.0	-12.1	28.2	-13.3	28.8	-14.5	29.3	-16.1	30.1
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	17.1	6.2	8.3	12.0	16.9	21.5	25.6	27.0	26.4	23.3	17.4	11.7	8.1	(6)
(7)		DBStd	8.46	5.79	5.04	5.04	4.37	3.60	2.47	2.01	2.29	3.06	4.47	4.75	5.18	(7)
(8)		HDD10.0	413	147	86	39	3	0	0	0	0	0	3	36	101	(8)
(9)		HDD18.3	1555	376	283	205	77	14	0	0	0	4	72	206	320	(9)
(10)		CDD10.0	2998	31	38	99	211	358	469	526	509	400	231	85	41	(10)
(11)		CDD18.3	1098	1	2	7	35	113	219	268	251	154	42	5	2	(11)
(12)		CDH23.3	10287	2	16	74	353	1087	2109	2663	2353	1238	349	36	6	(12)
(13)		CDH26.7	4349	0	1	8	84	407	961	1252	1076	474	85	2	0	(13)
(14)	Wind	WSAvg	2.9	2.9	3.2	3.5	3.5	3.2	2.8	2.6	2.3	2.7	2.6	2.7	2.7	(14)
(15)	Precipitation	PrecAvg	1157	87	83	94	76	77	118	125	130	126	84	77	74	(15)
(16)		PrecMax	1420	174	175	155	153	189	245	218	279	325	225	200	133	(16)
(17)		PrecMin	814	22	37	17	19	37	15	42	30	26	0	14	21	(17)
(18)		PrecStd	147	47	36	35	36	33	61	51	55	77	57	48	29	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	23.1	25.8	27.8	31.2	35.0	36.5	37.1	37.3	34.5	31.7	26.6	24.1	(19)
(20)			MCWB	17.7	18.2	18.0	20.2	22.3	23.6	25.0	25.2	22.8	23.2	19.5	19.0	(20)
(21)		2%	DB	20.9	22.4	25.5	28.9	32.4	34.5	35.3	34.9	32.9	28.8	24.0	21.8	(21)
(22)			MCWB	15.8	15.9	17.3	19.2	21.6	23.6	24.8	24.6	22.7	21.0	17.7	17.7	(22)
(23)		5%	DB	18.1	19.9	23.2	27.4	30.9	33.1	33.9	33.3	31.3	27.1	22.4	19.5	(23)
(24)			MCWB	14.4	14.3	15.5	18.4	21.2	23.3	24.7	24.6	22.8	19.9	17.0	16.6	(24)
(25)		10%	DB	16.1	17.6	21.1	25.4	28.9	31.9	32.5	32.1	29.6	25.1	20.3	17.4	(25)
(26)			MCWB	12.9	13.2	14.7	17.6	20.6	23.0	24.4	24.2	22.2	19.6	15.9	14.2	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	19.1	19.4	19.6	22.6	24.3	26.0	27.3	27.6	25.5	24.7	21.3	20.4	(27)
(28)			MCD B	21.5	23.6	24.2	27.1	29.4	32.1	33.6	32.6	30.5	27.6	23.8	22.6	(28)
(29)		2%	WB	17.2	17.8	18.6	21.1	23.1	25.3	26.4	26.3	24.7	23.4	19.7	18.7	(29)
(30)			MCD B	19.7	20.9	23.3	26.2	28.8	31.4	32.5	31.6	29.3	26.9	22.1	20.6	(30)
(31)		5%	WB	15.4	15.9	17.5	20.0	22.4	24.7	25.7	25.7	24.1	22.2	18.5	17.0	(31)
(32)			MCD B	17.5	18.8	20.8	24.9	28.0	30.2	31.5	30.9	28.4	25.0	20.9	19.0	(32)
(33)		10%	WB	13.2	13.8	16.3	18.9	21.7	24.1	25.2	25.1	23.5	20.8	16.9	14.9	(33)
(34)			MCD B	15.4	16.8	20.1	23.5	26.9	29.1	30.6	30.1	27.3	23.7	19.6	17.2	(34)
(35)	Mean Daily Temperature Range	MDBR		11.4	12.2	12.9	13.1	12.5	11.2	10.7	10.5	10.8	12.5	13.2	11.7	(35)
(36)		5% DB	MCD BR	13.6	14.8	15.2	15.0	14.4	13.1	12.6	12.4	12.9	13.9	14.5	13.0	(36)
(37)			MCWBR	9.3	9.3	7.8	6.8	5.3	4.2	4.0	4.1	4.6	6.7	8.7	9.3	(37)
(38)		5% WB	MCD BR	11.7	13.6	12.5	12.1	11.5	10.8	10.7	10.6	10.0	10.0	11.0	11.6	(38)
(39)			MCWBR	9.3	10.0	7.9	6.4	4.7	4.1	4.1	4.1	4.1	5.5	7.9	9.3	(39)
(40)	Clear-Sky Solar Irradiance	taub		0.319	0.330	0.367	0.403	0.463	0.492	0.539	0.513	0.429	0.393	0.339	0.314	(40)
(41)		taud		2.492	2.454	2.383	2.277	2.160	2.118	2.007	2.070	2.301	2.363	2.475	2.541	(41)
(42)		Ebn at Noon		888	916	905	883	831	804	763	775	831	831	855	873	(42)
(43)		Edh at Noon		84	97	114	133	151	158	175	162	122	106	85	76	(43)
(44)	All-Sky Solar Radiation	RadAvg		2.69	3.34	4.42	5.54	6.06	6.38	6.06	5.49	4.67	3.84	2.94	2.34	(44)
(45)		RadStd		0.16	0.36	0.34	0.38	0.48	0.43	0.38	0.26	0.39	0.48	0.24	0.20	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	+1.19	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (68 neighbors)		+0.49	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+168	+120

Nomenclature: See separate page